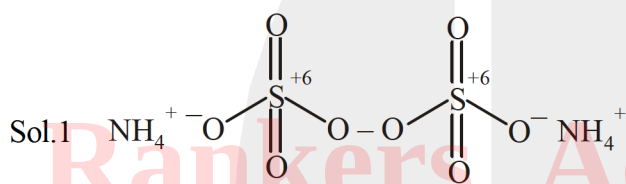
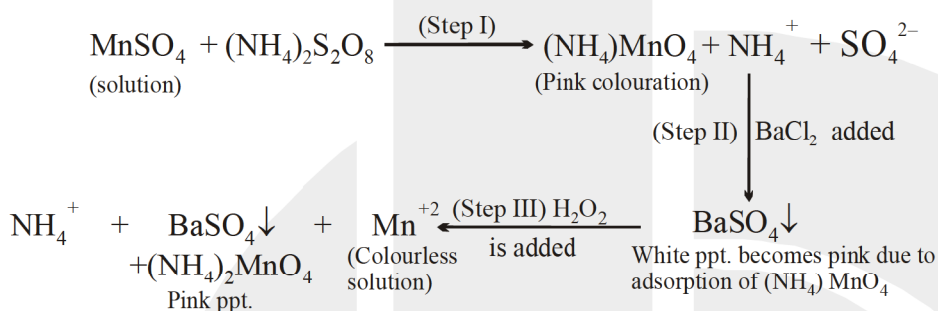


HINTS & SOLUTIONS

Q.1 (A)

Q.2 (B)

Q.3 (D)

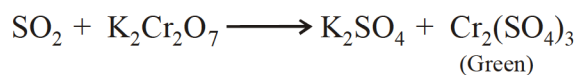
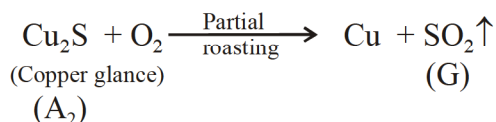
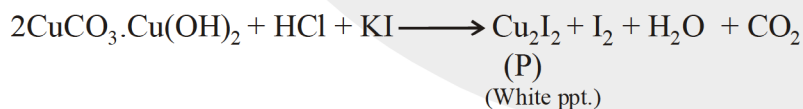
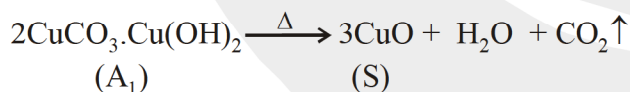


Sol.2 BaSO_4 is white but appears pink due to adsorption of $(\text{NH}_4)\text{MnO}_4$ on it.

Q.4 (D)

Q.5 (B)

Q.6 (A)



Q.7 (A)
Among C and CO; CO is present at a lower position in the Ellingham's diagram with respect to Fe below 710°C, thus CO will be a better reducing agent due more negative value of ΔG .

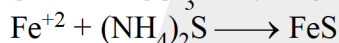
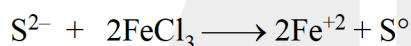
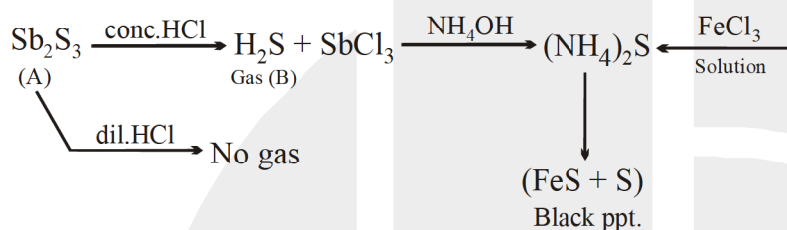
Q.8 (B)
All other reactions are non spontaneous at 1100°C due to +ve value of ΔG° .

Q.9 (A)
Silicon can reduce MgO but at very high temperature.

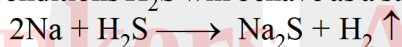
Q.10 (C)

Q.11 (B)

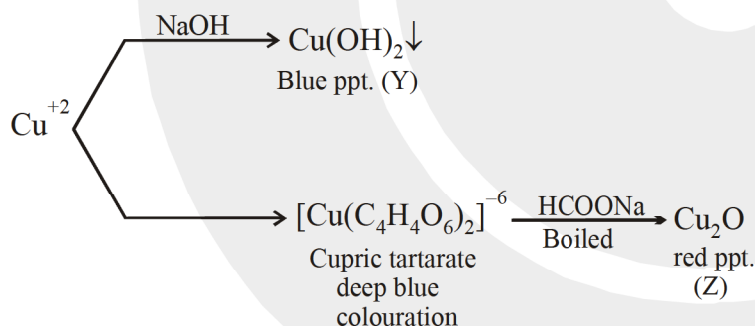
Q.12 (D)



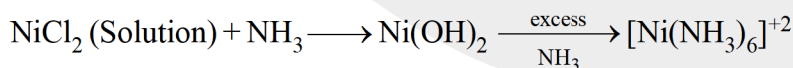
Sol.12 In hot conditions H_2S will behave as a strong acid and Na, Cu and Pb all will liberate H_2 as:



Q.13 (D)



Q.14 (A)



Q.15 (A)

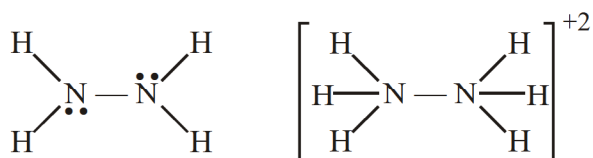
Galena : PbS ;

Chalcocite : Cu_2S ;

Limonite : $\text{FeO(OH).nH}_2\text{O}$

Azurite : $2\text{CuCO}_3 \cdot \text{Cu(OH)}_2$

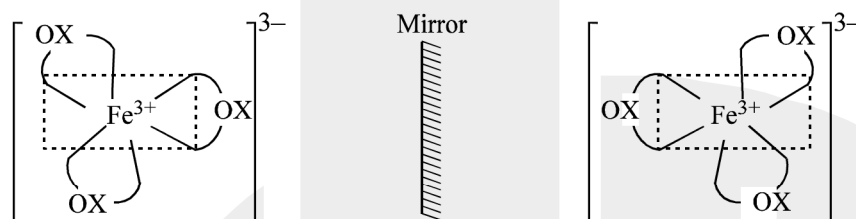
Q.16 (B)



Apply Bent's rule.

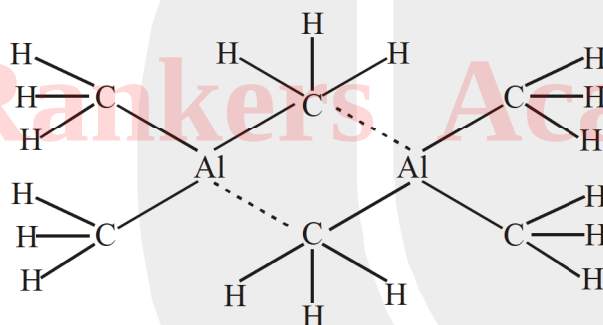
Q.17 (D)

Q.18 (B)



Configuration is asymmetrical, hence optically active.
Mabcd square planar Exhibits geometrical isomerism
Mabcd tetrahedral Complex is optically active]

Q.19 (C)



Q.20 (A)

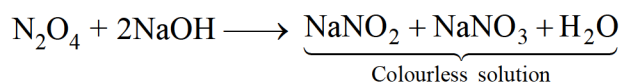
Q.21 (A)

Q.22 (BCD)

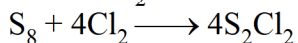
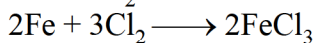
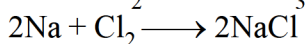
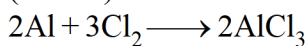
Q.23 (BD)

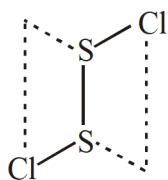
Group II reagent (H_2S) + Fe^{+3} / oxidising agent $\longrightarrow$ S (turbidity)

Q.24 (BC)

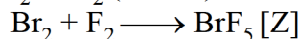
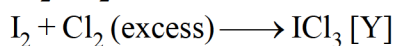
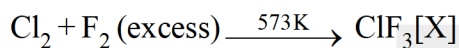


Q.25 (ABCD)





Q.26 (ABCD)



Q.27 (ABD)

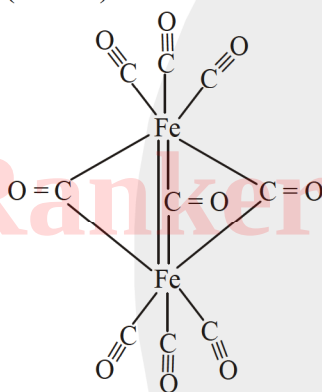
Small sized cations such as Li^+ forms covalent compound with F^- and thus it is difficult to dissociate.

Q.28 (BCD)

Mn_3O_4 , Fe_3O_4 , Co_3O_4 are mixed oxides.

Q.29 (ABCD)

Q.30 (ABCD)



Q.31 (A) QR (B) RS (C) S (D) PQ (E) S]

Q.32 (A) RM (B) SK (C) PL (D) QJ]

Q.33 (6)

Cu has highest electrode potential in 3d-series.

Q.34 (6)

Except Na_2ZnO_2 , all have either O_2^- or O_2^{2-} which have longer bond length than O_2^+ of $\text{O}_2[\text{PtF}_6]$]

Q.35 (6)

$[\text{Co}(\text{Ox})_3]^{3-}$, $[\text{Mn}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{gly})_3]$, $[\text{Pt}(\text{NH}_3)_6]^{4+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{IrCl}_6]^{3-}$ have d^2sp^3 hybridised central atom.